Frank–Kasper partitions in the diblock foam model

Exact certificates and the unresolved global comparison

Original numerical screen: Claude Opus 5.5 (effort xhigh).
Audit, additional certificates, analysis and manuscript: GPT-6-Astra (effort xhigh).
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Abstract. The diblock foam model balances interface area against the second moments of space-filling cells. An existing numerical screen of 121 catalogue Frank–Kasper topologies, with three additional comparators, finds its lowest value for the σ phase. We audit that screen and provide exact, reproducible geometric results. Integer certificates verify embedded tetrahedral realizations of all 121 catalogue entries. A second implementation checks six explicit foam partitions, their topology and their energy bounds, including rigorous transport to the intended crystallographic boxes. Five competitors have energies below the values printed in the source preprint; for a relaxed σ box we certify $\inf \mathcal{F} \leq 1.068575909$. We strengthen the spherical relaxation to the uniform lower bound $\mathcal{F} > 1.00000025$, using the Kepler sphere-packing theorem and exact rational constants; a self-contained elementary version gives $\mathcal{F} > 1.000000004$. We also certify global bounds for all affine images of the six stored meshes by reducing their deformation problem to convex minimization. Linear divergence calibrations give topology-wide lower bounds, certified for all 121 entries. A separate theorem puts a ceiling below 1.066 on this particular lower-bound construction, so it cannot separate competitors from the present σ witness. Eight additional exact witnesses improve the previously unrelaxed entries; one also disproves the necessity of the centroid-cone admissibility test. Neither these certificates nor additional symmetry-breaking restarts establish the ordering of the topology infima. We give complete proofs of the certified statements and identify the missing global inequality. This is a partial result, not a solution of the global minimization problem.

1 The precise question and its scope

Reddy et al.[1] compare 11 Frank–Kasper (FK) structures and body-centred cubic (BCC) partitions in the diblock foam model (DFM). Their numerical comparison favours σ. Extending this comparison to the catalogues of Dutour Sikirić, Delgado-Friedrichs and Deza[2] and Dutour Sikirić and Deza[3] is the question studied here. It is a question framed by this project; these sources do not formulate the catalogue-wide DFM minimization as an explicit open conjecture.

The catalogues contain 84 and 37 structures, respectively. Their enumeration is bounded by 20 and 21 cells in a reduced fundamental domain under a fixed-point-free crystallographic group. This number is generally smaller than the number of cells in the translation period used by the computations. The catalogue is not a classification of all FK structures.

An admissible partition consists of bounded cells with positive volumes, periodic under $L\mathbb{Z}^3$, with the prescribed face, edge and vertex incidences. Faces may be curved: this is part of the source definitions. Piecewise smooth and piecewise planar competitors are included. Cells have spherical boundaries, faces are disks, and the topology is kept fixed. Write $\mathcal{P}_T(L)$ for this class for topology T. With box relaxation, the quantity to

compare is

$$m_T = \inf_{\det L>0} \inf_{X\in\mathcal{P}_T(L)} \mathcal{F}(X).$$

Each topology uses the translation period represented by its input complex; arbitrary larger supercells are a separate enlargement of this problem. The desired conclusion would be $m_\sigma \le m_T$ for every competitor T. An optimizer returning a feasible X_T establishes only $m_T \le \mathcal{F}(X_T)$.

2 Energy identity and a universal lower bound

Let $V = \det L$, let n be the number of cells per period, and set $V_0 = V/n = 4\pi R_0^3/3$. For a lifted cell Ω_i, define

$$v_i = |\Omega_i|, \qquad c_i = v_i^{-1}\int_{\Omega_i} x\,dx, \qquad J_i = \int_{\Omega_i} |x-c_i|^2\,dx.$$

Let A denote the total interface area per period, counting each interface once, and let $J = \sum_i J_i$. Thus $\sum_i |\partial\Omega_i| = 2A$ and

$$\mathcal{A} = \frac{2A/n}{4\pi R_0^2}, \qquad \mathcal{I} = \frac{J/n}{4\pi R_0^5/5}, \qquad \mathcal{F} = (\mathcal{A}^2\mathcal{I})^{1/3}.$$

Substitution eliminates both n and π:

$$\boxed{\mathcal{F}^3 = \frac{20A^2J}{27V^3}.} \tag{1}$$

The expression is unchanged by translation, rotation or uniform rescaling.

Theorem 2. Every admissible partition satisfies $\mathcal{F} \ge 1$. In fact $\mathcal{F} > 1$ for a face-to-face periodic tiling by bounded cells with positive interface areas. This initial argument alone does not assert a uniform positive gap between an infimum and 1; the next section establishes such a gap.

Proof. The Euclidean isoperimetric inequality gives

$$A \ge \tfrac{1}{2}(36\pi)^{1/3}\sum_i v_i^{2/3}.$$

For completeness, the moment bound needs no convexity assumption. Fix the centroid c_i and let B_i be the ball of volume v_i centred there, with radius r_i. The sets $\Omega_i\backslash B_i$ and $B_i\backslash\Omega_i$ have equal measure. On the former $|x-c_i|^2 \ge r_i^2$, and on the latter $|x-c_i|^2 \le r_i^2$. Subtracting the two integrals proves

$$J_i \ge \int_{B_i} |x-c_i|^2\,dx = \frac{3}{5}\left(\frac{3}{4\pi}\right)^{2/3} v_i^{5/3}.$$

Equation (1) consequently yields

$$\mathcal{F}^3 \ge \frac{\left(\sum_i v_i^{2/3}\right)^2 \sum_i v_i^{5/3}}{V^3} \ge 1.$$

The last inequality is Hölder applied to $v_i = (v_i^{2/3})^{2/3}(v_i^{5/3})^{1/3}$ with exponents $3/2$ and 3. Equality with all $v_i > 0$ requires equal volumes and spherical cells. Two disjoint equal-volume balls cannot share a face of positive area, so they cannot form the stipulated

tiling. This excludes pointwise equality, but is not a compactness argument for a uniform gap. □

Proposition 3. For an isolated curved cell, its FK face-incidence graph does not improve the ball constants in Theorem 2. This cellwise relaxation cannot distinguish catalogue entries.

Proof. Each FK cell graph is the graph of a polyhedral sphere. Map this sphere homeomorphically onto a round sphere and regard its images of vertices, edges and faces as a subdivision of that round sphere. The resulting curved cell has precisely that face-incidence graph and achieves both ball constants. If nonzero dihedral features are required, arbitrarily small smooth or piecewise planar perturbations approach the same constants. For any list of cell types, the relaxation that forgets compatibility between neighbouring cells can therefore use equal-volume balls for all types. It has value 1. Compatibility between different cells is exactly what this relaxation has discarded. □

3 A uniform gap above 1

The gap below is intentionally modest. It controls all periods and all positive cell volumes, but is far too weak to order FK topology infima. Its purpose is to retain a quantitative part of the compatibility between cells that the isolated-ball relaxation loses.

Lemma 4. Let B be the ball of volume v and radius r centred at the centroid of a cell Ω, and put $t = |B\backslash\Omega|/v$. Then

$$J_\Omega - \frac{3}{5}\,vr^2 \;\geq\; \frac{2}{3}\,vr^2t^2.$$

Proof. The missing and excess sets both have volume tv. The smallest extra moment comes from adding the shell just outside B and removing the shell just inside its boundary. This follows directly by comparing $|x-c|^2$ on each side of the corresponding shell boundary. Thus the moment excess is at least

$$\frac{3}{5}\,vr^2\big((1+t)^{5/3} + (1-t)^{5/3} - 2\big).$$

The expression in parentheses has value and derivative zero at zero. Its second derivative is $\frac{10}{9}\left((1+t)^{-1/3} + (1-t)^{-1/3}\right) \geq 20/9$ for $0 \leq t < 1$, by convexity. Two integrations and continuity at $t = 1$ give the result. □

Lemma 5. Let $\rho < 1$ be an upper bound for the density of any periodic packing of equal balls in $\mathbb{R}^3$. Choose $0 < \eta, \lambda, \tau < 1$, and set

$$a = 1-\eta, \quad b = 1+\eta, \quad q = \frac{a(1-\lambda)}{b}, \quad C = \min\left\{\eta^2, \frac{10}{9}\,a^2\tau^2\right\}.$$

If $a^3\lambda^2(3-\lambda)/2 > 2\tau b^3$, every admissible partition obeys

$$\mathcal{F}^3 \geq 1 + C\left(1 - \frac{\rho}{q^3}\right). \tag{6}$$

Proof. For each cell let $r_i = (3v_i/(4\pi))^{1/3}$ and let t_i be as in Lemma 4. Write $\mathbb{E}f = V^{-1}\sum_i v_i f_i$. Uniformly rescale the partition so that $\mathbb{E}(1/r) = 1$. The area bound now gives $A \geq 3V/2$, hence

$$\mathcal{F}^3 \geq \frac{5J}{3V} \geq \mathbb{E}r^2 + \frac{10}{9}\,\mathbb{E}(r^2t^2).$$

Cauchy–Schwarz gives $\mathbb{E}r \geq 1$. Consequently, for $\delta = \mathcal{F}^3 - 1$,

$$\mathbb{E}\Big((r-1)^2 + \frac{10}{9}\,r^2t^2\Big) \leq \delta. \tag{7}$$

Call a cell good if $a \leq r_i \leq b$ and $t_i \leq \tau$. Every other cell contributes at least C per unit of its volume weight to the left-hand side. The good cells therefore occupy a fraction at least $1 - \delta/C$ of the volume. This estimate remains valid when that lower bound is negative.

Consider any two distinct good cells in the infinite periodic tiling, including different translates of the same representative. Their centroid balls B_i, B_j have intersection volume at most $t_iv_i + t_jv_j \leq 2\tau(4\pi/3)b^3$: the cells themselves are disjoint up to sets of measure zero. If their centres had distance at most $2a(1-\lambda)$, the radius-a balls at those centres would intersect in volume at least

$$\frac{4\pi}{3}\,a^3\,\frac{\lambda^2(3-\lambda)}{2}\,.$$

This is the volume of two spherical caps of height $a\lambda$. The assumed strict inequality excludes such a pair. Thus radius $R = a(1-\lambda)$ balls at all good centres form a periodic packing.

If there are m good representatives, their total cell volume is at most $m(4\pi/3)b^3$. The packing density is consequently at least $(1-\delta/C)(R/b)^3 = (1-\delta/C)q^3$ and at most ρ. Rearrangement proves (6). No lower bound on any individual cell volume or on the box aspect ratio was assumed. □

For a self-contained packing estimate, the classical Blichfeldt weight $f(x) = \max\{1 - |x|^2/2, 0\}$ gives $\rho \leq 5/(4\sqrt{2})$. Indeed, for unit-ball centres separated by at least 2, let m centres lie at distance less than $\sqrt{2}$ from a point x. If $m > 0$, then

$$\sum_{i=1}^{m} |c_i - x|^2 = \frac{1}{m}\sum_{i<j} |c_i - c_j|^2 + m|\bar{c} - x|^2 \geq 2(m-1).$$

It follows that $\sum_i f(x - c_i) \leq 1$; for $m = 0$ this is immediate. Integration over one period, with $\int_{\mathbb{R}^3} f = 16\pi\sqrt{2}/15$, proves the density bound. The historical reference is Blichfeldt [4]; the argument needed here has just been given in full.

Theorem 8. Every admissible partition satisfies $\mathcal{F} > 1.000000004$ by the elementary argument above. Using the established Kepler theorem, the stronger bound is $\mathcal{F} > 1.00000025$. Both constants are uniform in topology, number of cells per period, positive cell volumes and nonsingular box shape.

Proof. For the elementary version choose $(\eta, \lambda, \tau) = (3/4000, 1/32, 7/10000)$ and $u = 1.000000004$. Set $p = q^3(1 - (u^3 - 1)/C)$. Exact rational arithmetic verifies the strict lens condition, $p > 0$, and $p^2 > 25/32 = \rho^2$. Equation (6) then gives $\mathcal{F} > u$.

The Kepler theorem [5] gives $\rho = \pi/\sqrt{18}$. Choose $(\eta, \lambda, \tau) = (9/2500, 7/100, 7/2000)$ and $u = 1.00000025$. The same exact checks now give $p > 0$ and $p^2 > 242/441 > \pi^2/18$, where the latter inequality uses $\pi < 22/7$. For example, that elementary bound follows by integrating the positive function $x^4(1-x)^4/(1+x^2)$ on $(0,1)$: its integral is $22/7 - \pi$. The script `check_uniform_gap.py` verifies all rational constants and writes their strict margins. It does not reverify the Kepler theorem or formalize the geometric proof. □

Thus degenerating sequences cannot approach energy 1. This is not a compactness theorem near σ: the A15 vanishing-cell example below is compatible with the new bound, which is much smaller than 1.068575909.

4 Why the integer geometry certificates prove embeddedness

The following elementary degree statement makes the geometric verification precise. A periodic piecewise affine map is described by finitely many vertex representatives and integer lattice shifts.

Lemma 9. Let K be a closed connected oriented triangulated 3-manifold and let $f : K \to \mathbb{R}^3/L\mathbb{Z}^3$ be piecewise affine. Suppose its determinant on every oriented tetrahedron is strictly positive. If the sum of its tetrahedron volumes is $\det L$, then f is a homeomorphism.

Proof. Integrating the volume form gives $\deg(f) = 1$, since

$$\deg(f) \det L = \sum_{\tau} \operatorname{vol}(f(\tau)) = \det L.$$

A regular value outside the finite image of the two-dimensional skeleton has one preimage: all local orientation signs are positive.

To include the skeleton, the fibre over any point is finite, since f is injective on each tetrahedron. Isolate each preimage in a neighbourhood whose boundary does not map to the chosen value. Its local degree is positive: a sufficiently close regular value in the image of an incident tetrahedron has at least one positively oriented preimage in that neighbourhood, and every other contribution is nonnegative. The sum of these positive local degrees is 1. Thus every fibre is a singleton. The continuous bijection from a compact space to the torus is a homeomorphism. □

4.1 All 121 catalogue topologies

The certificate stores fractional vertex positions on the grid 2^{-44}. It lists the periodic tetrahedra of each catalogue contact complex. Crystallographic expansion and floating-point coordinates are used only to propose these data. The verifier uses integers for the following tests: every tetrahedron has nonzero volume; after positive orientation every triangle occurs twice with cancelling orientations; the vertex links are connected oriented triangulated 2-spheres; and the sum of determinants is exactly $6\,2^{132}$, the value for a unit-volume torus.

The face pairing and spherical links give a 3-manifold. The degree argument then verifies its embedded periodic realization, not merely a volume identity within a tolerance. Connectedness is also forced by positivity and total degree one: every closed component would have positive integer degree.

In each vertex link the verifier checks 12, 14, 15 or 16 vertices, all of degree 5 or 6, exactly twelve of degree 5, and no edge joining two degree-6 vertices. The dual cells therefore have the four FK types, with isolated hexagons. The barycentric dual of the certified triangulation is a genuine embedded foam with that topology.

Theorem 10. The 121 integer data sets in `certificates/catalogue_topology.npz` give embedded periodic FK complexes with the incidence structures extracted from the 84 part-I and 37 part-II catalogue files. All 121 exact checks pass.

This verifies the supplied realizations. The identification of the catalogue and its enumeration range comes from the source papers and archive[2] [3] [6], not from a new enumeration in this work. The manifest retains hashes of the CGD inputs. An entry that is not Delaunay at its supplied coordinates is not thereby proved incapable of another Voronoi realization.

4.2 The six energy witnesses

Each energy certificate instead stores the actual refined foam triangles, their shared vertex representatives, periodic shifts, cell incidences and a chosen interior apex for each cell. The audit checks every cell boundary as a connected oriented 2-sphere, including circular vertex links. Each coarse interface is a disk with one boundary circle. Periodic cell interfaces are unique, each foam edge meets three cells and faces, and each foam vertex meets four cells and six faces. The extracted labelled periodic dual tetrahedra agree exactly with the corresponding complex reconstructed from the crystallographic registry.

Coning the cell boundaries to their apices gives the requisite triangulated 3-manifold. Strict positivity of all cone determinants and the exact volume identity then allow Lemma 9 to be applied. Thus the triangles describe an embedded partition with the declared topology.

5 Exact integration and box correction

For an oriented boundary triangle with vertices a, b, c relative to its cell apex, let $d = \det(a, b, c) > 0$ and $w = d/6$. The contribution of its cone to volume, first moment and second moment is

$$v_\tau = w, \qquad M_\tau = \frac{w}{4}\,(a + b + c), \tag{11}$$

$$S_\tau = \frac{w}{20}\left(|a + b + c|^2 + |a|^2 + |b|^2 + |c|^2\right). \tag{12}$$

Indeed, for the three nonzero barycentric coordinates of a uniform tetrahedron, the averages are $1/4$, their squares have average $1/10$, and distinct products have average $1/20$. Expanding the integrands proves these formulas. After summing over a cell,

$$J_i = S_i - \frac{|M_i|^2}{v_i}\,.$$

All these quantities are rational on the integer grid. A triangle area equals $\sqrt{q}/2$, where q is the squared integer cross-product norm, with the grid scale understood. Integer floor and ceiling square roots enclose every area. Substitution into (1), followed by exact rational comparisons of cubes, encloses the energy of the particular mesh. The lower endpoint is *not* a lower bound on m_T.

Lemma 13. Let T be linear with $\|T - I\|_2 \le \varepsilon < 1$ and $\det T > 0$. For any partition X,

$$\mathcal{F}(TX) \le \frac{(1+\varepsilon)^2}{(1-\varepsilon)^2}\,\mathcal{F}(X).$$

Proof. Areas increase by at most $(1+\varepsilon)^2$. Centroids transform by T, so $J(TX) \le \det(T)(1+\varepsilon)^2 J(X)$, while $V(TX) = \det(T)V(X)$. Consequently (1) gives $\mathcal{F}(TX) \le (1+\varepsilon)^2 \det(T)^{-2/3}\mathcal{F}(X)$. Since $\det T \ge (1-\varepsilon)^3$, the claimed bound follows. □

For each fixed box the audit forms $T = B\widehat{L}^{-1}$, where $\widehat{L}$ is the stored dyadic box and B is the exact intended box. Arb ball arithmetic at 192-bit precision proves $\|T - I\|_F < 10^{-12}$. Rectangular aspect ratios are exact rationals; the hexagonal box uses $\sqrt{3}/2$. The Frobenius bound implies the operator norm bound in Lemma 13. The correction leaves all nine-decimal upper bounds unchanged.

Theorem 14. The six certificates give the following upper bounds on the appropriate topology infima. The first five are in the exact stated boxes; the last uses the exact relaxed dyadic box stored in its certificate.

Topology	Box convention	Certified upper bound	Source value
BCC	cubic	1.071107810	1.071111
P	$1.91 : 3.57 : 1$	1.068897166	1.068909
M	$1.89 : 1 : 3.3$ $(Pnma)$	1.069011740	1.069038
H	$1 : 3.88 : 1$	1.069208084	1.069219
μ	hexagonal, $c/a = 5.2$	1.069546495	1.069662
σ	relaxed dyadic box	1.068575909	–

Each of the first five bounds is also below the lower rounding endpoint of the corresponding printed six-decimal number.

The tabulated source values agree in arXiv:1806.01726v1 and the published PNAS SI Appendix, Figs. S3, S8, S11, S5 and S12. The SI was retrieved from Europe PMC; all twelve unconstrained DFM energies and caption box ratios were checked visually. For the hexagonal cases the source uses a doubled orthorhombic repeat. Repeating our hexagonal witness in that supercell preserves $\mathcal{F}$, since A, J, V all double in (1). These are improved feasible competitors, not a contradiction of a proved theorem. No unique cause of the numerical differences is established.

Combining Theorems 8 and 14 yields the global, but insufficiently narrow, enclosure

$$1.00000025 \le \min_{T \text{ in the catalogue}} m_T \le 1.068575909. \tag{15}$$

The same enclosure holds upon adding P, M and BCC.

6 Global affine minimization of a fixed partition

One part of the optimization does admit a convex reduction. Here the entire cell geometry is fixed before one applies a common affine map. This does not include independent changes of faces or cell volumes.

Let $S = \sum_i \int_{\Omega_i} (x - c_i)(x - c_i)^{\mathsf{T}} dx$ be the centred moment tensor of a fixed partition X, with period volume V. It is positive definite. Integrate once over each interface, and write n for its unit normal; its sign is immaterial. For $G \succ 0$ define

$$a(G) = \int \sqrt{n^{\mathsf{T}} G^{-1} n}\, dA, \qquad j(G) = \operatorname{tr}(GS).$$

If $T^{\mathsf{T}} T = G$, area transforms by $|\det T|\, |T^{-\mathsf{T}} n|$, the moment tensor by $|\det T| TST^{\mathsf{T}}$, and volume by $|\det T|$. All determinant factors cancel from the energy identity:

$$\mathcal{F}(TX)^3 = \frac{20}{27V^3}\, a(G)^2 j(G). \tag{16}$$

Theorem 17. The minimization of $\mathcal{F}(TX)$ over nonsingular affine maps is equivalent to minimizing the convex function $a(G)$ subject to $G \succ 0$ and $\operatorname{tr}(GS) = j_0$, for any fixed $j_0 > 0$. At a chosen $G \succ 0$ put $a = a(G)$, $j = j(G)$ and

$$K = \int \frac{G^{-1} n n^{\mathsf{T}} G^{-1}}{\sqrt{n^{\mathsf{T}} G^{-1} n}}\, dA.$$

If $\kappa S - K$ is positive semidefinite and $b = (3a - \kappa j)/2 > 0$, then every nonsingular affine map satisfies

$$\mathcal{F}(TX)^3 \geq \frac{20 b^2 j}{27V^3}\,. \tag{18}$$

Proof. For fixed n, the identity

$$\sqrt{n^{\mathsf{T}} G^{-1} n} = \sup_{z \neq 0,\ n^{\mathsf{T}} z \geq 0} \frac{n^{\mathsf{T}} z}{\sqrt{z^{\mathsf{T}} G z}}$$

is Cauchy–Schwarz in the G metric, with equality at $z = G^{-1} n$. Each function in the supremum is convex in G, since $u \mapsto u^{-1/2}$ is convex for $u > 0$. Integration proves convexity of a. Scaling G multiplies a by the inverse square root and j by the scale, leaving $a^2 j$ unchanged. This proves the normalization claim.

Differentiation gives $\nabla a(G) = -K/2$, and $\operatorname{tr}(KG) = a$. For any $H \succ 0$ normalized by $\operatorname{tr}(HS) = j$, the supporting-plane inequality gives

$$a(H) \geq a - \tfrac{1}{2} \operatorname{tr}(K(H - G)) = \tfrac{3}{2}\, a - \tfrac{1}{2} \operatorname{tr}(KH) \geq \tfrac{1}{2}\, (3a - \kappa j) = b.$$

The last step follows from $\kappa S - K \succeq 0$ and $H \succ 0$. Equation (16) completes the global bound. □

6.1 Six certified affine families

The script `check_affine_orbits.py` integrates all six components of S by exact rational cone formulas. If $d > 0$ is the determinant of a tetrahedral cone with apex at zero, other vertices u, v, w, and $s = u + v + w$, its second-moment matrix is

$$\frac{d}{120}\, (ss^{\mathsf{T}} + uu^{\mathsf{T}} + vv^{\mathsf{T}} + ww^{\mathsf{T}}).$$

After summing per cell, subtract the outer product of its first moment divided by its volume. This also recovers the scalar moments of the previous energy audit.

A floating-point convex optimizer proposes a metric, which is rounded to an exact dyadic matrix. It is not part of the proof. Outward ball arithmetic at 192 bits verifies positive definiteness of G, S and $\kappa S - K$ by Sylvester's criterion, evaluates (18), and checks rational decimal endpoints. The saved rational metrics can be checked without running the optimizer again. Hashes identify the meshes covered by the earlier topology audit.

Stored partition	Lower bound on affine infimum	Upper bound
BCC	1.071107807	1.071107811
P	1.068892909	1.068892956
M	1.068962667	1.068962682
H	1.068948492	1.068948499
μ	1.069099093	1.069099122
σ	1.068575891	1.068575910

In particular, every affine image of each of the five stored competitor meshes lies above the existing σ witness. This is a comparison of six affine families, not of six topology classes. Internal relaxation of the cells after a box change enlarges those families and may lower their infima; indeed the inherited box-relaxed H run is below the affine lower bound for the stored fixed-box H mesh. There is no contradiction.

7 Topology-wide divergence bounds and their limitation

Unlike the affine families above, the next bound allows arbitrary internal cell deformations. Label a contact edge by (i, j, s) if cell i in the base period meets cell j translated by Ls, with $s \in \mathbb{Z}^3$. Let W be a real 3×3 matrix of positive determinant and let $z_1, \dots, z_n \in \mathbb{R}^3$. Suppose that

$$|z_i - z_j + Ws| \le 1 \qquad \text{for every contact edge } (i, j, s). \tag{19}$$

These conditions use only the periodic topology.

Theorem 20. For every admissible partition with these incidences, arbitrary positive cell volumes and any box L of positive determinant,

$$A \ge 3(\det W)^{1/3} V^{2/3}, \qquad \mathcal{F} \ge \left(\frac{6 \det W}{\pi n}\right)^{2/9}. \tag{21}$$

Proof. Set $B = WL^{-1}$ and, on the lifted cell indexed by (i, k), use the field $\xi_{i,k}(x) = B(x - Lk) + z_i$. Its divergence is $\operatorname{tr} B$. On a shared interface the difference of the two fields is $z_i - z_j + Ws$, independent of position. The divergence theorem and (19) therefore give $V \operatorname{tr} B \le A$. This argument applies to curved as well as planar faces.

Replacing (W, z_i) by (RW, Rz_i) for any proper rotation R preserves (19). The maximum of $\operatorname{tr}(RWL^{-1})$ over these rotations is the sum of the three singular values of WL^{-1}: its determinant is positive, so the maximizing orthogonal factor can be chosen proper. Their arithmetic–geometric mean is at least $3(\det W/V)^{1/3}$, proving the area bound.

Ball rearrangement and Jensen's inequality give, without requiring equal cell volumes,

$$J \geq \frac{3}{5}\left(\frac{3}{4\pi}\right)^{2/3}\sum_i v_i^{5/3} \geq \frac{3}{5}\left(\frac{3}{4\pi}\right)^{2/3} V^{5/3} n^{-2/3}.$$

Substitute this and the area bound into (1). Cubing the resulting inequality for $\mathcal{F}^3$ yields $\mathcal{F}^9 \geq 36(\det W)^2/(\pi^2 n^2)$, as claimed. □

Proposals maximize $\log\det W$ under (19), with W symmetric positive definite and $z_1 = 0$. This is a convex optimization problem; symmetry of W loses no positive-determinant candidates, by a common left rotation (polar decomposition). Solver status is not a proof. For each saved proposal we round W, z_i to integer arrays and divide them by one integer larger than the square root of the maximum squared edge norm. The verifier checks every edge inequality with integers, checks $\det W > 0$, and uses 192-bit outward arithmetic for π and the final decimal bound. It also rechecks the exact catalogue complex. Thus verification requires no optimizer, external source archive or assumption of numerical optimality.

All 121 certificates pass; 34 improve the uniform bound 1.00000025. Selected certified topology-wide lower bounds are

$$m_{\mathrm{A15}} \geq 1.008666655, \qquad m_H \geq 1.008808885,$$
$$m_\sigma \geq 1.004288649, \qquad m_{\mathrm{E11}} \geq 1.004321474.$$

There is a structural obstruction to obtaining the desired separating bound by improving this optimization alone.

Proposition 22. If the periodic contact triangulation has t tetrahedra and n vertices, every feasible pair (W, z) in (19) satisfies

$$\det W \leq \frac{t}{6\sqrt{2}}, \qquad \left(\frac{6\det W}{\pi n}\right)^{2/9} \leq \left(\frac{t^2}{2\pi^2 n^2}\right)^{1/9}.$$

For all 121 catalogue entries the rightmost number is less than 1.066. This is a ceiling on the bound in (21), *not an upper bound on any physical energy or topology infimum.*

Proof. Map the contact vertex (i, k) to $-z_i + Wk$, and extend affinely over each contact tetrahedron. Every mapped edge has length at most one by (19). The map is equivariant from the original torus to the torus with period matrix W. Its induced lattice map is the identity in these coordinates; it is homotopic to the corresponding linear map, by straight interpolation of its periodic displacement. Hence it has degree one. Equivalently, cancellation of the oriented periodic faces gives the signed-volume identity

$$\sum_{\text{oriented contact tetrahedra}} \mathrm{Vol}_{\mathrm{signed}}(\text{image}) = \det W.$$

The map may fold or collapse some tetrahedra; neither possibility affects this identity. For each saved certificate the verifier additionally checks the identity by exact integer determinants.

For completeness, a tetrahedron with all six edges of length at most one has volume at most $1/(6\sqrt{2})$. Indeed, for its vertices $q_0, \dots, q_3$ and their mean $\bar{q}$, put $S = \sum_a (q_a -$

$\bar{q})(q_a - \bar{q})^{\mathsf{T}}$. Then

$$\operatorname{tr} S = \tfrac{1}{4}\sum_{a<b}|q_a - q_b|^2 \le \tfrac{3}{2}\,, \qquad \mathrm{Vol}^2 = \frac{\det S}{9} \le \frac{(\operatorname{tr} S/3)^3}{9} \le \frac{1}{72}\,.$$

Taking absolute values in the signed-volume sum proves the determinant bound. Direct counts in the separately rechecked catalogue give $t/n \le 124/21$, with equality for part-II Y04–Y06. Therefore the method's ceiling is at most

$$\left(\frac{(124/21)^2}{2\pi^2}\right)^{1/9} = 1.0652530398\ldots < 1.066.$$

The final strict inequality is checked with outward arithmetic. □

These calibrations supply valid topology-dependent information over the full admissible classes, but cannot reach the present witness level 1.068575909. A stronger method must use information discarded by this separate area and ball-moment estimate, or a different lower certificate.

8 Numerical evidence and its limits

The inherited screen uses triangulated interfaces, midpoint refinements, analytic gradients and restricted normal-motion L-BFGS cycles, with tangential smoothing. It includes homogeneous box strains. Its leading Richardson-extrapolated energies are as follows; these estimates have *no certified lower error endpoints*.

Topology	Extrapolated $\mathcal{F}$	Relative gap from σ
σ	1.0685748	0
E11	1.0688088	0.0219%
H	1.0688092	0.0219%
μ	1.0688388	0.0247%
K_7Cs_6 type	1.0688396	0.0248%
N56	1.0688683	0.0275%
P	1.0688742	0.0281%

The full historical table contains 124 entries. Of the 121 catalogue structures, 105 have their supplied contact triangulation realized as a Delaunay triangulation; 16 do not at those coordinates. Several of the latter relaxations collapse an edge or fail the sufficient centroid-cone test. Eight historical entries retain only valid unrelaxed starting values; the auxiliary-apex continuation below now supplies improved certified witnesses for all eight. Those topologies have not been energetically excluded. Finite mesh values calculated in floating point should not be described as exact numerical certificates merely because the underlying integration formulas are exact.

The validation script recomputes the flat-input checks and equal-volume Kelvin and Weaire–Phelan examples, and compares the stored fixed-box data against the source tables. It finishes with zero failed checks. Agreement on stored data is distinguished from recomputing every high-resolution relaxation: most historical checkpoint files were absent from this checkout.

Additional restarts perturb all vertex coordinates by independent Gaussian noise of scale $0.015R_0$, and the box by symmetric random strains of scale about 0.01. No crystallographic symmetry is imposed. Two fixed seeds, 1701 and 2903, were run through refinement levels 0–2 for σ, E11, H and μ. All 24 completed stages pass the floating-point cone test. The largest difference between the two level-2 results of the same structure is 1.6×10^{-9}; the ordering remains unchanged. This evidence concerns only the tested starts and resolutions.

8.1 A valid cell need not be star-shaped about its centroid

A new continuation uses a moving auxiliary apex for each cell. For the oriented determinant d of each boundary cone, a logarithmic barrier $-\log(d - d_0)$ enforces positivity, with d_0 chosen to correspond to 10^{-7} times the mean cell volume. The energy always uses the true volume centroid; the auxiliary apex serves only to certify geometry. Six barrier weights, $10^{-1}, \ldots, 10^{-6}$, are used at mesh level zero, with up to 5000 L-BFGS iterations per weight and a freely strained box. Some stages reach their iteration limit with non-negligible gradients. They provide trial partitions, not stationary or optimal solutions. This search uses cells star-shaped about some auxiliary point, which is still only a sufficient subclass of the admissible curved cells.

The part-II Y04 example supplies a precise correction to the use of the older centroid-cone diagnostic. Its saved dyadic partition has 42 cells, the exact prescribed dual complex, positive auxiliary cone determinants and an exact degree-one volume sum. Its energy is at most 1.254466833. Nevertheless, 184 boundary cones based at the *true volume centroids* have strictly negative determinants, verified using rational first moments. Thus the partition is embedded although the centroid-cone test fails. Positivity at the centroid is a sufficient condition under the other manifold hypotheses, not a necessary condition for admissibility. Failure of that diagnostic alone cannot justify discarding a topology or calling its cells self-intersecting.

All eight previously unrelaxed entries now have additional exact witnesses:

Topology	Certified upper bound	Negative centroid cones
N55	1.102285105	0
N59	1.101112653	0
N60	1.129410812	0
II:Y02	1.103468432	0
II:Y03	1.103507483	0
II:Y04	1.254466833	184
II:Y05	1.254006234	370
II:Y06	1.255816821	281

The verifier checks the exact periodic identity directly against the standalone catalogue, the manifold hypotheses, every positive auxiliary cone, the exact volume sum and rational energy bounds. The 48 numerical continuation stages are retained separately, including stopping reasons and gradient norms. These improved upper bounds do not exclude a lower infimum for any of the eight topology classes.

9 The missing global step

A certified ordering would follow from a feasible σ energy U and bounds L_T such that $\mathcal{F}(X) \geq L_T > U$ for *every* admissible X of every other topology. No such bounds are supplied by the screen. Refinement and restarts do not supply them either. A positive Hessian at a particular stationary mesh would only be a local statement, and would still require a continuum error argument.

The obstruction is not missing optimization software. It is control over all admissible curved faces, all positive cell volumes and all allowed box shapes, including degenerating sequences. A fixed-topology class need not be closed under such degenerations, so an attained minimizer cannot be presumed.

9.1 A certified restricted family and a finite degeneration

The one-parameter flat-face A15 family in the source SI, Eq. (S16) (preprint Eq. (A16)), has energy

$$F_c^3 = \frac{5}{55296}\,(a+bc^2)^2(3c^4-5c^3+10), \qquad a = 12+12\sqrt{6}, \quad b = 6\sqrt{5}-4\sqrt{6}-3 > 0.$$

Here the cubic period has side length 2, and the two Z_{12} cells have volume $c^3/2$ each. The other six cells have volume $(4-c^3/2)/3$. The parameter range on which this formula describes the family includes positive c approaching zero and the sample $c = 1.22$ used below.

Proposition 23. For every $c \geq 0$ the displayed algebraic expression satisfies $F_c > 1.0699$. Moreover $F_{1.22} < 1.0701$. Thus no admissible member of this particular flat A15 family improves the certified σ witness. This is not a lower bound for unrestricted curved A15 cells.

Proof. Let $P(c) = 5(a+bc^2)^2(3c^4-5c^3+10) - 55296(1.0699)^3$. On $[0,2]$ we verify positivity by Bernstein coefficients, using outward ball arithmetic with 192-bit precision for the radicals. The six intervals have endpoints

$$0, \quad 1, \quad 9/8, \quad 19/16, \quad 5/4, \quad 3/2, \quad 2.$$

For clarity, the complete conversion rule is as follows. If $P(c) = \sum_{j=0}^{8} p_j c^j$, then on $[l, l+h]$ set $q_i = h^i \sum_{j=i}^{8} \binom{j}{i} p_j l^{j-i}$. Its Bernstein coefficients are $\beta_k = \sum_{i=0}^{k} q_i \binom{k}{i} / \binom{8}{i}$. The verifier proves every $\beta_k > 0$ on each interval. Since the Bernstein basis is nonnegative and sums to one, $P > 0$ there.

For $c \geq 2$, the derivative of $(a+bc^2)^2(3c^4-5c^3+10)$ equals

$$c(a+bc^2)\,[bc^3(24c-35) + 3ac(4c-5) + 40b] > 0.$$

The already checked value at 2 completes the half-line argument. Direct outward evaluation at the rational sample $61/50$ proves the upper bound. All steps are implemented in `scripts/check_a15_family.py`; no sampled minimization is used. □

As $c \downarrow 0$, the two Z_{12} cells disappear, but

$$F_c \longrightarrow \left(\frac{50(12+12\sqrt{6})^2}{55296}\right)^{1/3} = 1.157131807984\ldots.$$

Thus bounded energy alone does not prevent vanishing cell volumes even within an FK family. This particular limit is above the σ witness; it does not settle other degenerating families or show that degenerations can occur near the observed optimum. Any exclusion of those limits needs an additional argument.

9.2 Necessary stationarity is not global minimality

For a smooth interface oriented outward from cell i toward cell j, let $h = \operatorname{div}_S n$ be the sum of principal curvatures. A normal displacement u has first variations

$$\delta A = \int hu \, dS, \qquad \delta J = \int (|x - c_i|^2 - |x - c_j|^2) u \, dS.$$

The dependence of the centroids causes no additional term, since $\int_{\Omega_i} (x - c_i) \, dx = 0$. Thus an interior stationary partition in a fixed box must satisfy

$$2Jh + A(|x - c_i|^2 - |x - c_j|^2) = 0$$

on each smooth interface. Triple-line variations additionally give the usual equal-tension force balance. These are necessary conditions, not a convexity or uniqueness theorem.

9.3 A sufficient form of a global lower certificate

Normalize $V = 1$ by scaling. For positive A, J,

$$\mathcal{F} = \frac{20^{1/3}}{9} \inf_{s>0} (2sA + s^{-2} J), \tag{24}$$

because the optimizing value satisfies $s^3 = J/A$. This infimum identity warns against bounding just one chosen linear combination: a lower bound is needed uniformly in s.

Here is one possible sufficient mechanism. Index all periodic copies of the cells by α, with centres c_α, and choose continuously differentiable vector fields ξ_α equivariant under lattice translation. Suppose that for every pair and every position $|\xi_\alpha - \xi_\beta| \le 2s$. Summing the divergence theorem over the cells in a period gives

$$2sA + s^{-2} J \ge \int_{\mathbb{R}^3/L\mathbb{Z}^3} \inf_{\alpha} \Big(\operatorname{div} \xi_\alpha(x) + s^{-2} |x - c_\alpha|^2 \Big) dx.$$

At a shared interface this follows from $(\xi_\alpha - \xi_\beta) \cdot n \le 2s$; the remaining inequality is pointwise. A uniform lower bound $9U/20^{1/3}$ on the right for all s, centres and allowed boxes would prove $\mathcal{F} \ge U$ through (24). No fields attaining such a bound near the observed σ value have been constructed here. Restrictions to a fixed set of centres would leave the original question unresolved.

10 Reproduction and verification status

From the problem directory, with Python 3.12 or later:

```
python scripts/audit_certificates.py
python scripts/certify_catalogue.py
python scripts/test_audit_certificates.py
python scripts/check_a15_family.py
python scripts/check_uniform_gap.py
python scripts/check_affine_orbits.py
python scripts/test_affine_orbits.py
python scripts/check_topology_calibrations.py
python scripts/test_topology_calibrations.py
python scripts/check_barrier_witnesses.py
python scripts/test_barrier.py
python scripts/check_milestone1.py
python scripts/check_restarts.py
```

The standalone catalogue certificate needs no third-party archive at verification time. Regeneration uses `certify_catalogue.py --build`, after restoring the Zenodo archive with MD5 `582371cee665c62e01a2f47a73dd2c9d`. The manifest records its input filenames and individual SHA-256 hashes. Source PDFs remain local and are excluded from redistribution by the repository ignore rules.

The energy audit has a separate exact integration implementation from the optimizer and the original certificate program. It also verifies manifold links, interface topology and affine transport. Its regression checks reject reversed triangles, missing tetrahedra, duplicated faces and a pinched surface, and check invariance under translation and scaling. These are implementation checks, not a formal proof-assistant verification.

The topology-calibration verifier additionally checks all periodic edge norms and mapped signed-volume identities using integers. Four tests cover an analytic cubic torus, gauge translations, false norm and energy claims, and a missing tetrahedron. Three barrier tests check a directional derivative, translation invariance, boundary rejection and the independence of the true centroid calculation from the auxiliary apex.

The affine-orbit tests additionally check the exact tensor of a cube, its transformation under a non-diagonal shear and translation, the known affine minimum of the cube, and rejection of false dual certificates.

This continuation used a single agent, as requested. The new code and mathematical arguments have not undergone a fresh review. The global ordering $m_\sigma \le m_T$ remains unproved; the numerical screen, the exact feasibility statements and the global enclosure (15) are the conclusions supported by the present work.

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